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Research article

A 4-year field measurement of N_2O emissions from a maize-wheat rotation system as influenced by partial organic substitution for synthetic fertilizer

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ABSTRACT

Soil N_2O emissions depend on the status of stoichiometric balance between organic C and inorganic N. As a beneficial management practice to sustain soil fertility and crop productivity, partial substitution of organic fertilizers (OFs) for synthetic fertilizers (SFs) can directly affect this balance status and regulate N_2O emissions. However, no multi-year field studies of N_2O emissions under different ratios of OF_S to SFs have been performed. We conducted a 4-year experiment to measure N_2O emissions in a maize-wheat rotation in central China. Six treatments were included: total SF (TS), total OF, no N fertilizer, and ratios of to SF with 1: 2 (LO), 1: 1 (MO), and 2: 1 (HO), based on N content. Two incubation experiments were performed to further interpret the field data. In the first year, cumulative N_2O emissions (kg N ha^{-1}) in LO, MO, and HO were 4.59, 4.68, and 3.59, respectively, significantly lower than in TS (6.67). However, from the second year onwards, organic substitution did not reduce N_2O emissions and even significantly enhanced them in the fourth year relative to TS. Soil respiration under OF-amended soils increased over the course of the experiment. From the second year onwards, there was no marked difference in mineral N concentrations between OF- and SF-amended soils. OF caused a drop in soil pH. Cumulative N_2O was negatively correlated with pH. Long-term organic substitution enhanced N_2O emissions produced via denitrification rather than nitrification and resulted in higher temperature sensitivity of N_2O emissions than TS. The enhanced N_2O emissions from the OF-treated soils were mainly attributable to accelerated OF decomposition, increased denitrification-N_2O emissions, and lessened N_2O reduction due to lower pH and greater NO_3^-. These results indicate that OF substitution can reduce N_2O emissions in the first year, but in the long-term it can increase emissions, especially as soils warm.

1. Introduction

As one of the most important non-CO_2 greenhouse gases, nitrous oxide (N_2O) has a global warming potential 298 times higher than that of CO_2 at the 100-year time horizon (IPCC et al., 2007). It is also the single most important culprit causing destruction of stratospheric ozone (Ravishankara et al., 2009). The N_2O concentration in the atmosphere has been growing dramatically from pre-industrial value of about 270 ppb–330 ppb at present (Meinshausen et al., 2017), mainly attributed to anthropogenic reactive nitrogen (Nr) input (Gruber and Galloway, 2008). Agricultural soil receives the major part (75%) of the Nr (Galloway et al., 2004) and contributes more than 50% of total global anthropogenic N_2O emissions (Hu et al., 2015). It is therefore important to understand the impacts of Nr on N_2O emissions from agricultural soil.

Synthetic and organic fertilizers are the two main forms of anthropogenic Nr to agricultural soil (Gu et al., 2015). Synthetic fertilizers (SFs) have higher nutrient contents than organic ones, but they easily result in environmental pollution and soil degradation (Laird et al., 2010; Ahmed et al., 2017). Although organic fertilizers (OFs) are more eco-friendly, they are too low in nutrient concentration and slow in nutrient release to support crop production in a short time (Aguilera et al., 2013; Xiao et al., 2017). Partial substitution of organic fertilizers for synthetic ones has proven to be a beneficial approach to sustain soil fertility and crop productivity with comparison of applying synthetic or organic fertilizers alone. N_2O emissions mainly result from microbial processes of autotrophic nitrification and heterotrophic denitrification, which depend strongly on multiple factors, including soil substrate properties and environmental conditions (Bouwman, 1996; Dalal et al.,

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2003; Charles et al., 2017). Partial substitution of synthetic fertilizers by organic ones can influence N_2O emissions by directly mediating availability of soil carbon (C) and inorganic N substrates (Charles et al., 2017; Gómez-Muñoz et al., 2017; Zhang et al., 2018). Furthermore, it may also affect N_2O production by its indirect impact on soil pH, oxygen partial pressure and aeration (Xu et al., 2008; Cui et al., 2016; Zhang et al., 2018). Mechanisms through which partial organic substitution affects N_2O emissions are rather complex.

To date, complete substitution of synthetic fertilizers by organic ones has been shown to significantly increase, decrease and have no impact on N_2O emissions (Aguilera et al., 2013; Charles et al., 2017; Escanhoela et al., 2019; Zhou et al., 2017; Zhang et al., 2018). There are at least six avenues by which organic fertilizer substitutions influence N_2O emissions. First, organic amendments provide more bioavailable organic C compounds as energy for denitrifying microbes, consequently enhancing N_2O production (Charles et al., 2017). Second, addition of organic fertilizer can markedly enhance denitrification enzyme activity and promote denitrification (Fan et al., 2014). Third, a more anoxic condition created by decomposition of organic substrates facilitates denitrification (Cui et al., 2016). Fourth, the organic amendments with great C: N ratios may reduce the supply of inorganic N substrates to denitrifying or nitrifying microbes due to N immobilization, which may consequently decrease N_2O emissions (Charles et al., 2017). Fifth, compared with synthetic fertilizers, organic fertilizers avoid a concentrated release of mineral N and therefore mitigate N_2O emissions (Meijide et al., 2007). Sixth, organic amendments can enhance electron flow driving denitrification, which further promote the reduction of N_2O to N_2, especially in soils with low NO_3^- concentration (Senbayram et al., 2012). These theories indicate that organic fertilizer substitutions can promote denitrification and N_2O emissions by increasing substrate C supply and providing a more anoxic environment. However, as opposed to synthetic fertilizers, organic fertilizers may also decrease N_2O emissions by reducing inorganic N concentrations and promoting the reduction of N_2O. Either way, the status of stoichiometric balance between organic C and inorganic N is thought to heavily affect N_2O emissions. Unlike complete substitution, partial substitution of synthetic fertilizers by organic ones can provide both organic C and inorganic N, directly altering proportions of soil organic C and inorganic N, thereby regulating N_2O emissions. Therefore, optimizing the ratio of organic residue inputs to inorganic fertilizer N is likely an important tool for farmers to reduce N_2O emissions. So far, however, no multi-year agronomic field studies of N_2O emissions under different ratios of organic to synthetic fertilizers have been performed. Moreover, it is not clear how nitrification and denitrification respond to the different ratios.

Changes in temperature may alter the effect the ratio of organic to synthetic fertilizer has on N_2O emissions. It is because the changes can affect not only activity of denitrifying or nitrifying bacteria, but also substrate C supply and decomposition (Cui et al., 2016; Yin et al., 2017). In general, there are some inter-annual variations in air temperature during the growing season in a region. The variations may become stronger due to global climate change (Zhang and Huang, 2012). Therefore, it is necessary to estimate the response of N_2O emissions from the soils treated with different ratios of organic to synthetic fertilizer to the temperature variations. But the related information has not been reported.

In this study, we conducted 4 years of field measurement of soil N_2O emissions from a maize–wheat rotation system in the Huai River Basin, which accounts for 10% of the cultivated land in China and provides 20% of the total agricultural production (Sun et al., 2017). In addition, two independent incubation experiments were carried out to determine contributions of nitrification and denitrification to N_2O production and effect of temperature change on N_2O emissions, respectively. The objectives of this study were to: 1) characterize and quantify multi-year N_2O emissions under different supply ratios of organic to synthetic fertilizer, and 2) compare the differences in N_2O emission between treatments and years, and identify the main reason for the differences. We hypothesized that 1) compared with the application of synthetic fertilizer alone, a suitable replacement of organic fertilizer might reduce N_2O emissions, and 2) the response of N_2O emissions to temperature might be different under various supply ratios of organic to synthetic fertilizer.

2. Material and methods

2.1. Site description and experimental design

The field study was conducted at the Nongcui Experimental Station (NES) of Anhui Agricultural University (31°52′ N, 117°25′ E), on the Huai River Basin, China. This region lies in the northern edge of the subtropical zone, with an average annual precipitation of 998 mm and a mean air temperature of 15.7 °C. The soil of this experiment field is classified as eutic planosol (FAO classification). Prior to the start of the experiment, the topsoil contained 1.36 g kg^{-1} N and 12.64 g kg^{-1} organic C. The contents of total phosphorus (P) and potassium (K) were 0.67 g kg^{-1} and 18.0 g kg^{-1}, respectively. The soil pH was 7.56.

In the field experiment, we used the typical double cropping system in the Huai River Basin, with winter wheat and summer maize in rotation. Winter wheat (Yangmai 13) was sown in mid to late October and harvested in late May; summer maize (Zhengdan 958) was sown in early June and harvested in late September. The study was performed over four consecutive cropping years from October 2014 to October 2018.

The field setup was a completely randomized block design with three replicates for each of six treatments. Each plot was 2.4 m × 6.0 m. According to N content, different ratios of synthetic and organic fertilizer and a control were used in six different treatments. Specifically, the treatments were: (1) no N fertilizer (CK), (2) total synthetic fertilizer (TS), (3) low ratio of organic to synthetic fertilizer (LO, organic N: synthetic N = 1 : 2), (4) medium ratio of organic to synthetic fertilizer (MO, organic N: synthetic N = 1 : 1), (5) high ratio of organic to synthetic fertilizer (HO, organic N: synthetic N = 2 : 1), and (6) total organic fertilizer (TO). The synthetic N and organic fertilizers applied were urea and rapeseed cake, respectively. The rapeseed cake had a C/N ratio of 8.2 and a pH of 4.9. Apart from the CK, the N fertilizer application rates in all the treatments for both wheat and maize production were 240 kg N ha^{-1}. The P and K fertilizer application rates were 65.5 kg P ha^{-1} and 99.6 kg K ha^{-1} for wheat production, and 46.8 kg P ha^{-1} and 93.4 kg K ha^{-1} for maize production, respectively. The amounts of P and K fertilizers applied were the same in all of the treatments. The detailed fertilization information is shown in Table S1. The synthetic P and K fertilizers applied were superphosphate and potassium chloride, respectively. Before tilling, organic fertilizer and synthetic P and K fertilizers were applied as base fertilizer at one time. Synthetic N fertilizer was required to be applied two times in a crop growing season. A part of N fertilizer was applied as basal and the other part as supplemental (topdressing) at the jointing stage of wheat and the bell-mouthed stages of maize. Basal and supplemental N fertilizers were of the ratios of 4:1 and 7:3 for wheat and maize, respectively.

2.2. Measurements of N_2O flux

Soil N_2O fluxes were measured in situ using static chamber and gas chromatography techniques (Zhang et al., 2012), over a 4-year period from October 2014 to October 2018. In the center of each experimental plot, a stainless steel base frame (0.5 m × 0.3 m) was permanently inserting to a depth of 10 cm for the entire wheat and maize growing seasons. An opaque PVC chamber (0.5 × 0.3 × 0.5 m) with a digital thermometer was placed onto the installed base frames to collect gas samples. The chamber was coated on the outside heat-isolation material. The groove of base frame was filled with water to ensure the gas-tight seal. Gas fluxes were measured once a week, but the frequency was increased to twice weekly after each fertilizer application. Gas sampling was performed between 09:00 and 11:00 a.m. Four air samples were withdrawn from chamber headspace at an interval of 6 min for measuring N_2O concentrations, using 60-mL gas-tight plastic syringes. The gas samples were analyzed for N_2O concentrations within 8 h after sampling by a gas chromatograph (Agilent 7890 A, Agilent Ltd, America) equipped with an electron capture detector (ECD) at 300 °C. The carrier gas was argon-methane (5%) with a flow rate of 40 mL min^{-1}. The column temperature was maintained at 40 °C. N_2O flux was calculated from a linear increase in gas concentrations in the chamber headspace over time (Hutchinson and Livingston, 1993; Zou et al., 2005).

2.3. Soil sampling and two laboratory incubation experiments

We conducted soil sampling prior to basal fertilizer application in the wheat and maize growing seasons (Oct. 23, 2017 and Jun. 9, 2018). Six soil core samples (3.5 cm in diameter by 20 cm in length) were collected from each plot randomly and then mix thoroughly to form a single composite sample. Soil samples were homogenized by passing through a 4.0 mm mesh sieve and stored at 4 °C for further processes within 40 d.

In the first incubation experiment, an acetylene inhibition technique was used to distinguish between nitrification and denitrification as sources of soil N_2O (Maag and Vinther, 1996). Briefly, portions of 28 g fresh soil were placed into 120 mL serum vials and NH_4Cl solution was added to give a moisture content of about 70% of field capacity. The amendments corresponded to 50 mg NH_4^+-N per kg of dry soil. After serum vials were sealed with air-tight septa, acetylene (10 Pa) was added with a syringe to the vial headspace. Meanwhile, a control serum vial without acetylene was prepared for each sample. Based on the real field temperature, soils sampled in the wheat and maize planting seasons were incubated at 15 °C and 29 °C, respectively. After incubation for 0, 2, 4, 6, 8, and 11 days, a 5 mL of gas sample for each vial was collected for analyzing N_2O concentration using a gas chromatography (Agilent 7890a, Agilent Ltd, America). N_2O in the C_2H_2-treated vials was assumed to be produced by denitrification. Therefore, nitrification-N_2O could be calculated by the difference in N_2O generated between control and C_2H_2-treated vials (Davidson et al., 1986).

The second incubation experiment with different temperatures was performed to reveal the influence of inter-annual variations in temperature on soil N_2O emissions. According to the soil temperature at the peak of N_2O emission after basal fertilizer application during the 4-year experiment, we set the incubation temperature of soil samples in the wheat planting season at 13 °C, 14.5 °C and 17 °C and in the maize planting season at 24 °C, 25 °C, 26 °C and 29 °C, respectively. Other incubation conditions of soil samples, such as amount of N added to soil, soil humidity, sampling days and the estimation method of N_2O emissions, were exactly the same as the control treatment in the first incubation experiment above.

2.4. Auxiliary measurements

When gas samples were collected, the air temperature inside the chamber was recorded with a portable digital thermometer (JM222, Jinming Instrument Co. Ltd., Tianjin, China). Soil temperatures and moisture (at 5 cm depth) were monitored with ECH2O loggers (EM50 Data logger, Decagon Devices, WA). Water filled pore space (WFPS) was calculated based on the equation: WFPS% = volumetric moisture/(1-bulk density/2.65) × 100%. Precipitation was monitored by an automatic weather station. Soil NH_4^+-N and NO_3^--N contents were extracted by 1.0 M KCl and determined with an Auto Discrete Analyser (Clever-Chem380, DeChem-Tech, Germany). The gas samples taken to determine N_2O flux static chamber were also used to measure CO_2 concentrations with a gas chromatograph with TCD detector (Agilent 7890 A, Agilent Ltd, America). The carrier gas was N_2 (99.999%) at a flow rate of 23 mL min^{-1}. The TCD and column temperature were maintained at 250 °C and 40 °C, respectively. The calculation method of CO_2 flux was the same as that of N_2O flux.

At the end of four years of the field experiment, soil samples were taken at the depth of 0–20 cm for soil property analyses. Soil pH was measured with pH meter under a soil: water ratio of 1: 2.5. Soil organic C was determined using the $K_2Cr_2O_7$ oxidation method (Nelson and Sommers, 1996). Total N content in soil was measured using Kjeldahl Analyser (KT260, FOSS-SCINO, Denmark). Total P content was determined by molybdenum antimony spectrophoto-metric method, after digesting samples with H_2SO_4–$HClO_4$. Available P content was measured by 0.5 M $NaHCO_3$ extraction. A flame photometric method (FP6400, Jingke, China) was employed to determine total K after HF–$HClO_4$ digestion and available K with NH_4OAc extraction (Lu, 1999).

2.5. Statistical analysis

Analyses of variance (ANOVA), correlation and stepwise regression analysis were performed using the IBM SPSS Statistics software 19.0 ($p < 0.05$). Differences between treatments were evaluated using the least significant difference (LSD) test at a level of significance of 0.05. A two-way multivariate analysis of variance was applied to analyze the effects of year and fertilization on cumulative N_2O emissions, as well as temperature and fertilization on N_2O emission rates. All data presented were means ± standard errors (SE). The correlation between soil characteristics and cumulative N_2O emissions was identified by principal component analysis (PCA) using the Canoco for Windows 4.5 software.

3. Result

3.1. N_2O emission

Over a 4-year measurement period, the significant N_2O emission peaks appeared after the basal rather than supplemental fertilizer application (Fig. 1). The average cumulative N_2O emissions caused by basal fertilization accounted for 52.63–73.98% of annual N_2O emission across the fertilization treatments. There were great inter-annual differences in the size of the N_2O peaks between years. The highest N_2O peaks in the wheat growing season were observed on November 19, 2016, ranged from 947.1 to 1809.5 μg N m^{-2} h^{-1} across the experimental fertilization treatments. For the maize growing season, the highest N_2O fluxes were identified on June 28, 2015, with peak values varied from 459.9 to 1543.2 μg N m^{-2} h^{-1}.

The average cumulative N_2O emissions during the wheat and maize seasons accounted for 20.24–63.74% and 36.26–79.76% of the total annual emissions across the fertilization treatments, respectively (Table 1). Both planting year and fertilization had significant impacts on cumulative emissions of N_2O (Table S2). The interaction of planting year and fertilization was also significant ($p < 0.001$). During the 2014/2015, the cumulative emission of N_2O (kg N ha^{-1}) in the TS treatment was 6.67, which was significantly higher than those in the LO (4.59), MO (4.68), and HO (3.59) treatments (Table 1). However, during the 2015/2016, there was no significant difference between all treatments but CK. During the 2016/2017, the cumulative N_2O emissions in the OF but LO treatments were markedly higher than that in the TS treatment. During the 2017–2018, the cumulative N_2O emissions in all the OF treatments were remarkably greater than that in the TS treatment. Throughout the experimental period, no significant difference in average cumulative annual N_2O emissions and emission factor was observed between the fertilization treatments except the TO.

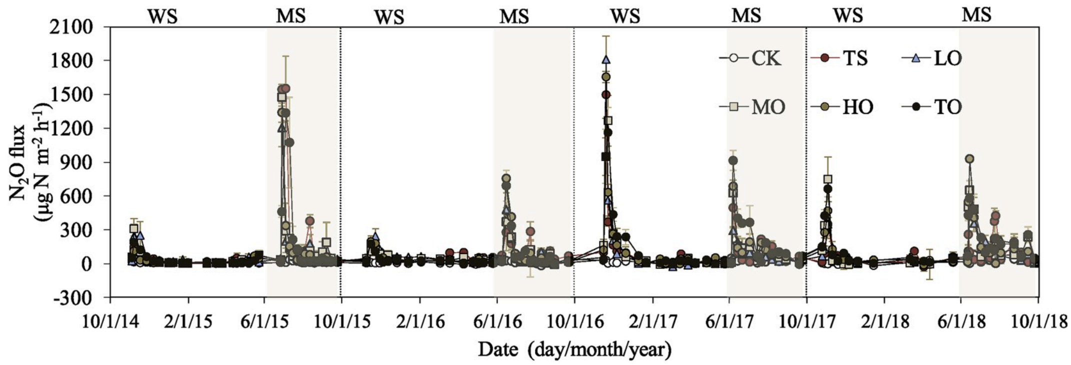

Fig. 1. Seasonal dynamics in N_2O flux under different treatments from Oct.2015 to Oct. 2018. CK, no N fertilizer; TS, total synthetic N; LO, organic N: synthetic N = 1 : 2; MO, organic N: synthetic N = 1 : 1; HO, organic N: synthetic N = 2 : 1; TO, total organic fertilizer. WS, wheat season; MS, maize season.

3.2. Environmental variables and related influence on N_2O emissions

The total precipitation was 952.8, 996.4, 948.7 and 1257.9 mm during the 2014/2015, 2015/2016, 2016/2017 and 2017/2018 maize-wheat rotations, respectively (Fig. 2a). The precipitation in the maize growing season accounted for 46.2–60.4% of the total amount over each year. The temperature and WFPS of the topsoil were shown in Fig. 2a and b, respectively. Soil temperatures corresponding to the N_2O peaks induced by basal fertilization were 13.1, 13.4, 17.0 and 14.5 °C for wheat season and 23.8, 26.0, 24.9 and 28.7 °C for maize season in the 2014/2015, 2015/2016, 2016/2017 and 2017/2018, respectively. The corresponding soil WFPS (%) were 30.8, 34.7, 36.0 and 45.7 for wheat season and 45.8, 28.2, 34.5 and 33.8 for maize season in the 2014/2015, 2015/2016, 2016/2017 and 2017/2018, respectively.

Topsoil mineral N release peaks occurred primarily following the basal and supplemental fertilization for different fertilization treatments (Fig. 3a and b). Across the fertilization treatments, the mean NH_4^+ concentrations varied from 7.76 to 9.04 mg N kg^{-1}, while mean NO_3^- concentrations were in the range of 22.07–27.43 mg N kg^{-1} from 2015 to 2018 (Fig. 3). There was no significant difference in mean NH_4^+ and NO_3^- concentrations or release rates between the fertilization treatments. The notable CO_2 emission peaks mainly occurred after the application of base fertilizer throughout the experimental period (Fig. 3c). With increasing cropping years, the sizes of the CO_2 peaks under OF treatments in both the wheat or maize seasons showed an increasing trend. Moreover, the size distribution became narrower. The average CO_2 fluxes in the OF treatments pronouncedly greater than that in the TS treatment during the whole experimental period.

Stepwise regression analysis was applied to identify the relationship between soil N_2O fluxes and environmental factors (Table 2). The result showed in the control, CO_2 fluxes were positively correlated with soil temperature. In the TS treatment, NH_4^+ concentration was a key factor affecting N_2O flux. Across the OF treatments, CO_2 fluxes were positively correlated with N_2O fluxes. Furthermore, NO_3^- concentration was also a main factor influencing N_2O fluxes.

3.3. Soil characteristics and correlation with cumulative N_2O emissions

After the 4-year experimental period, soil chemical characteristics were measured (Table 3). Soil pH (7.3) in the TS treatments was higher than those (6.5–6.7) in the LO, MO, and HO treatments. Organic C in the TS treatment were significantly lower than those in the OF treatments. Soil organic N also showed marked difference between the treatments and the highest value was found in the TO treatment. There was no significant difference in total P, total K, available P and available K between the treatments. A principal component analysis was used to identify the relationship between soil characteristics and cumulative emissions of N_2O (Fig. 4). The first principal component of PCA accounted for 89.5% of the total variation, while the second component explained about 6.1% of the total variation. The results showed that the annual accumulation of N_2O had a significantly positive correlation with soil organic C and N (r = 0.862 and 0.596, $p < 0.01$), while a pronouncedly negative relationship with soil pH (r = −0.694, $p < 0.01$). After four years of the maize-wheat rotation, the soil characteristics and cumulative N_2O emissions in the TS treatment had shown obvious difference with those in the OF treatments.

3.4. N_2O emission by nitrification and denitrification, and N_2O response to temperature

N_2O emission rates from nitrification and denitrification as determined by the first laboratory experiment are shown in Table 4. N_2O emission rates by denitrification from both the wheat- and maize-seasonal soils in the OF treatments were significantly higher than that in the TS treatment (Table 4). For wheat-seasonal soil, N_2O emission rates by denitrification in the HO and TO treatments were greater than those in the LO and MO treatments. For maize-seasonal soil, N_2O emission rate via denitrification in the TO treatment was the highest and there was no significant difference between the LO, MO, and HO treatments. Unlike denitrification, no significant difference in N_2O emission rate by nitrification was observed between the TS and OF treatments from the wheat- or maize-seasonal soils (Table 4). Soil pH had a significantly negative correlation with N_2O emissions produced by denitrification rather than nitrification (Fig. S1).

Response of N_2O emission rates to temperature fluctuations was determined by the second laboratory experiment, which is displayed in Table 5. N_2O emission rates of wheat-seasonal soil presented a significant increasing trend with increasing soil temperature (Table 5). However, N_2O emission rates of maize-seasonal soil did not show a significant change in response to temperature fluctuations until the temperature was 29 °C (Table 5). N_2O increase rates per degree rise in temperature showed pronounced difference between the treatments (Table 5). For wheat-seasonal soil, the N_2O increase rate in response to temperature rise in the TF treatment was lower than those in the MO and TO treatments, but higher than that in the HO treatment. For maize-seasonal soil, the N_2O increase rates in organic fertilizer treatments were greater than that in the TF treatment. The highest N_2O increase rates were observed in the HO treatment, more than triple that in TS.

Table 1
Cumulative emissions of N_2O (kg N ha^{-1}) and direct N_2O emission factors (EFd, %) under different treatments in the maize–wheat rotation system throughout the experimental period of 2014–2018.

Year/Treatment	Wheat season		Maize season		Annual		
	N_2O	S/A (%)	N_2O	S/A (%)	N_2O	EF_d	B/A (%)
2014–2015							
CK	0.49 ± 0.13 b	59.76	0.33 ± 0.01 d	40.24	0.82 ± 0.14 d	–	39.30 ± 3.63 c
TS	1.35 ± 0.24 ab	20.24	5.32 ± 0.47a	79.76	6.67 ± 0.30 a	2.43 ± 0.18 a	72.29 ± 3.23 b
LO	1.79 ± 0.56a	39.00	2.80 ± 0.54c	61.00	4.59 ± 1.09 bc	1.57 ± 0.40 ab	67.92 ± 2.35 b
MO	1.36 ± 0.34 ab	29.06	3.32 ± 0.56bc	70.94	4.68 ± 0.44 bc	1.61 ± 0.13 ab	69.01 ± 9.37 b
HO	0.80 ± 0.06 b	22.28	2.79 ± 0.79c	77.72	3.59 ± 0.83 c	1.15 ± 0.36 b	70.24 ± 0.79 b
TO	1.19 ± 0.18 ab	20.41	4.64 ± 0.62 ab	79.59	5.83 ± 0.62 ab	2.09 ± 0.26 a	86.50 ± 1.59 a
2015–2016							
CK	0.35 ± 0.06 d	53.85	0.30 ± 0.14c	46.15	0.65 ± 0.09 b	–	50.23 ± 5.98c
TS	2.50 ± 0.15 ab	55.93	1.97 ± 0.29 ab	44.07	4.47 ± 0.33 a	1.59 ± 0.14 a	57.12 ± 15.2 bc
LO	2.67 ± 0.07a	56.57	2.05 ± 0.37 ab	43.03	4.72 ± 0.41 a	1.69 ± 0.14 a	61.77 ± 6.66 ab
MO	2.15 ± 0.32bc	55.27	1.73 ± 0.24 b	44.73	3.89 ± 0.24 a	1.35 ± 0.07 a	56.27 ± 5.15 bc
HO	1.62 ± 0.35bc	38.21	2.62 ± 0.32a	61.79	4.24 ± 0.66 a	1.49 ± 0.27 a	75.72 ± 2.14 a
TO	1.39 ± 0.15c	36.48	2.42 ± 0.13 ab	63.52	3.81 ± 0.18 a	1.31 ± 0.11 a	70.95 ± 3.61 ab
2016–2017							
CK	0.43 ± 0.22c	32.09	0.91 ± 0.11c	67.91	1.34 ± 0.32 d	–	30.76 ± 1.91 d
TS	3.43 ± 0.17 b	60.92	2.21 ± 0.15bc	39.08	5.63 ± 0.29 c	1.79 ± 0.25 cd	64.72 ± 4.40 c
LO	3.41 ± 0.31 b	63.74	1.94 ± 0.19c	36.26	5.35 ± 0.33 c	1.67 ± 0.02 d	79.50 ± 3.77 ab
MO	4.23 ± 0.57 ab	59.16	2.92 ± 0.51bc	40.84	7.15 ± 0.55 b	2.42 ± 0.29 bc	74.17 ± 2.33 abc
HO	4.25 ± 0.19 ab	56.97	3.21 ± 0.45 b	43.03	7.46 ± 0.62 b	2.55 ± 0.22 b	69.29 ± 7.60 bc
TO	5.30 ± 0.52a	51.26	5.04 ± 0.38a	48.74	10.34 ± 0.62 a	3.75 ± 0.18 a	82.50 ± 2.24 a
2017–2018							
CK	0.44 ± 0.07c	27.50	1.16 ± 0.36c	72.50	1.60 ± 0.43 d	–	24.86 ± 4.07 b
TS	1.76 ± 0.29 b	39.73	2.67 ± 0.42bc	60.27	4.43 ± 0.51 c	1.18 ± 0.35 b	32.25 ± 5.84 c
LO	2.49 ± 0.17a	36.56	4.31 ± 0.29 ab	63.44	6.81 ± 0.43 ab	2.17 ± 0.34 ab	47.84 ± 3.71 b
MO	2.34 ± 0.25 ab	36.68	4.04 ± 0.29 ab	63.32	6.38 ± 0.46 b	1.99 ± 0.35 ab	62.51 ± 4.83 a
HO	2.55 ± 0.16 a	37.56	4.25 ± 0.60 ab	62.44	6.79 ± 0.60 ab	2.17 ± 0.29 ab	62.35 ± 2.59 a
TO	3.03 ± 0.11a	36.59	5.26 ± 0.50a	63.41	8.28 ± 0.60 a	2.79 ± 0.28 a	55.95 ± 3.30 a
Mean 2014–2018							
CK	0.43 ± 0.07 c	39.09	0.67 ± 0.14 c	60.91	1.10 ± 0.18 c	–	36.29 ± 1.30 d
TS	2.26 ± 0.07 b	42.64	3.04 ± 0.23 b	57.36	5.30 ± 0.27 b	1.75 ± 0.17 b	52.63 ± 2.15 c
LO	2.59 ± 0.17 ab	48.23	2.78 ± 0.05 b	51.77	5.37 ± 0.16 b	1.78 ± 0.13 b	64.26 ± 1.78 b
MO	2.52 ± 0.10 ab	45.65	3.00 ± 0.04 b	54.35	5.52 ± 0.14 b	1.84 ± 0.13 b	65.49 ± 1.87 b
HO	2.31 ± 0.13 b	41.77	3.22 ± 0.31 b	58.23	5.52 ± 0.40 b	1.84 ± 0.10 b	69.40 ± 2.21 ab
TO	2.73 ± 0.10 a	38.61	4.34 ± 0.18 a	61.39	7.06 ± 0.24 a	2.48 ± 0.04 a	73.98 ± 0.86 a

CK, no N fertilizer; TS, total synthetic N; LO, organic N: synthetic N = 1 : 2; MO, organic N: synthetic N = 1 : 1; HO, organic N: synthetic N = 2 : 1; TO, total organic fertilizer. S/A = ratio of seasonal/annual cumulative N_2O emission. B/A = ratio of N_2O accumulation caused by basal fertilizer application to annual N_2O accumulation. Different letters mean significant difference at $p < 0.05$ between treatments.

4. Discussion

4.1. Influence of synthetic and organic fertilizers on N_2O emission

Our results found that in the first year of the field experiment, partial substitution of SFs by OFs especially under the 67% organic substitution clearly reduced N_2O emissions compared with application of SF alone (Table 1), supporting our first hypothesis. This result is similar to the previous study that the substitution of 75% N through OF had the higher potential than synthetic N fertilizer alone to mitigate N_2O emissions (Hou et al., 2018). However, from the second year onwards partial organic substitution did not reduce cumulative N_2O emissions, and even

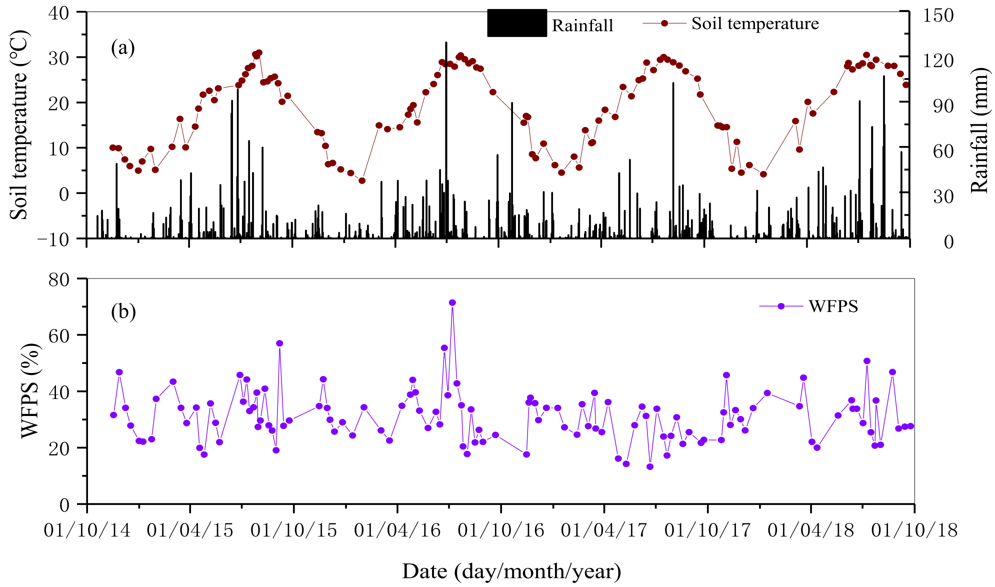

Fig. 2. Seasonal variations in (a) daily precipitation, soil temperature and (b) water filled pore space (WFPS %) during the gas flux measurement period.

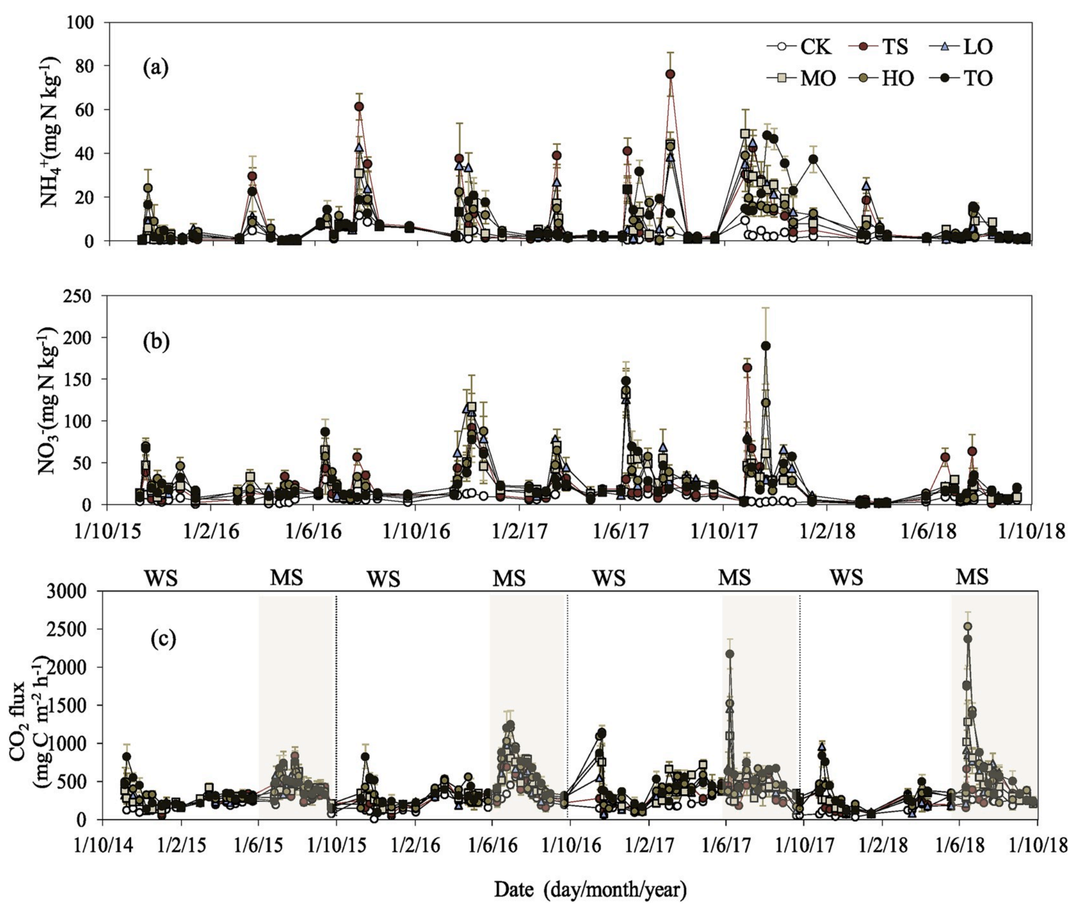

Fig. 3. Seasonal variations in ammonium (a) and nitrate (b) concentrations in surface soil and CO_2 flux (c) and under different treatments. CK, no N fertilizer; TS, total synthetic N; LO, organic N: synthetic N = 1 : 2; MO, organic N: synthetic N = 1 : 1; HO, organic N: synthetic N = 2 : 1; TO, total organic fertilizer. WS, wheat season; MS, maize season.

significantly enhanced them in the fourth year relative to the SF application alone (Table 1). The result contradicts our first hypothesis. There were at least two reasons why cumulative N_2O emissions was enhanced from soils treated by OF from the second year onwards.

First, unlike SFs, the application of can improve soil microbial biomass and activity by increasing energy availability (Xiao et al., 2007). The enhanced microbial activity adversely speeds up the decomposition of organic substance with a C/N ratio lower than 25/1

Table 2
Stepwise regression analysis for the relationship between soil N_2O fluxes and environmental factors.

Treatment	Multiple linear regression model	Model R^2	F-value	p-value
CK	$FN_2O = 0.779 + 1.512\ T^{**} - 0.048\ FCO_2^{*}$	0.317	16.019	0.001
TS	$FN_2O = 74.084 + 2.275\ CNH_4^{+*}$	0.078	5.898	0.018
LO	$FN_2O = -26.794 + 0.341\ FCO_2^{**}$	0.141	11.454	0.001
MO	$FN_2O = -58.194 + 0.251\ FCO_2^{**} + 2.661\ CNO_3^{-**}$	0.297	14.587	< 0.001
HO	$FN_2O = -111.436 + 0.339\ FCO_2^{**} + 1.887\ CNO_3^{-*}$	0.350	18.549	< 0.001
TO	$FN_2O = -91.429 + 0.342\ FCO_2^{**} + 1.980\ CNO_3^{-**}$	0.513	36.308	< 0.001

CK, no N fertilizer; TS, total synthetic N; LO, organic N: synthetic N = 1 : 2; MO, organic N: synthetic N = 1 : 1; HO, organic N: synthetic N = 2 : 1; TO, total organic fertilizer. FN_2O (μg N m^{-2} h^{-1}) and FCO_2 (mg C m^{-2} h^{-1}): instantaneous fluxes. CNH_4^+ (mg N kg^{-1}) and CNO_3^- (mg N kg^{-1}): concentrations of ammonium and nitrate in soil. T: soil temperature (°C). * and **: significant correlation at $p < 0.05$ and 0.01, respectively.

(Leite et al., 2017). In this study, the C/N ratio of organic fertilizer was only 8.2/1, far less than 25/1, facilitating OF decomposition. Soil respiration is a result of decomposition. We found that soil respiration under the OF treatments continued to increase annually (Fig. 3c). Therefore, the capability of slow-release N for OF may be gradually weakened, and the available N is released faster. Actually, we observed that mineral N release in soils treated by OF was not markedly slower than that in the SF-treated soils from the second year onwards (Fig. 3a and b). Therefore, OF might fail to mitigate N_2O emissions by avoiding a concentrated release of mineral N with time.

Second, the application of could gradually enhance N_2O emissions produced via denitrification with time. In our case, we observed N_2O emission rates from denitrification in the soils amended with OF was significantly higher than that in the SF-treated soil under both wheat and maize seasons in the fourth year of this study (Table 4). Further regression analyses also revealed that NO_3^- concentration, a primary substrate for denitrification, was a key factor influencing N_2O flux in the OF-treated soils (Table 2). The results indicate that in the manure-amended soils, denitrification plays an important role in N_2O emissions and the increase in N_2O emissions from denitrification may be another reason for the enhanced cumulative N_2O emissions. These results agree with previous studies that manure stimulated N_2O emissions via denitrification (Ding et al., 2007; Chantigny et al., 2010), but do not support the study by Kramer et al. (2006), showing that OF could not enhance N_2O emissions due to the increase in the capacity of N_2O reduction to N_2. In general, the capacity for N_2O reduction depends strongly on soil NO_3^- contents. Only when the concentration of NO_3^- is lower than 20 mg N kg^{-1} dry soil, can organic amendments effectively reduce N_2O to N_2 (Senbayram et al., 2012). In this study, the annual average NO_3^- concentrations across fertilization treatments ranged from 22.07 to 27.43 mg N kg^{-1} dry soil from the second year onwards, more than the threshold value. Therefore, it is difficult for OF to effectively

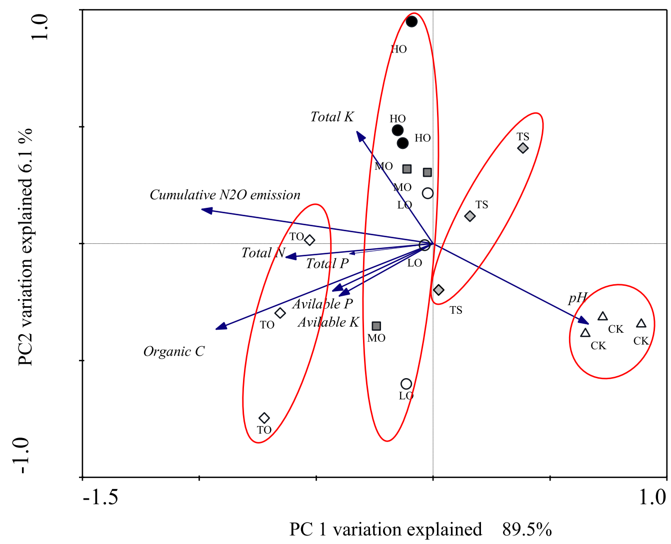

Fig. 4. Principal component analysis of cumulative emissions of N_2O and soil characteristics. CK, no N fertilizer; TS, total synthetic N; LO, organic N: synthetic N = 1 : 2; MO, organic N: synthetic N = 1 : 1; HO, organic N: synthetic N = 2 : 1; TO, total organic fertilizer.

reduce N_2O. Since OF promotes N_2O production rather than reduction, N_2O emissions can be enhanced. In addition, a decrease in soil pH caused by OF may also reduce the capacity for N_2O reduction. It is generally believed that OF can increase soil pH (Whalen et al., 2000), contrary to our result (Table 3). The main reason for the pH decrease could be that the applied OF was acidic (pH = 4.9). We found a negative correlation between soil pH and cumulative N_2O emissions (Fig. 4). This result agrees with the findings of Wang et al. (2018), but does not support the study by Park et al. (2018), showing that cumulative N_2O emissions were 23.3% lower in the pH 5 soil compared to the pH 7 soil. The discrepancy may be related to different response of N_2O production and consumption to the extent of pH decline. Further correlation analysis indicated that soil pH was negatively correlated N_2O emissions produced via denitrification rather than nitrification (Fig. S1). Previous studies have found that in denitrification, N_2O consumption is inhibited when soil pH is lower than 6.8, while N_2O production is inhibited only when it is less than 5.6 (Hénault et al., 2019; Song et al., 2019). In this study, partial organic substitution changed soil pH from 7.3 to 6.5, thus only inhibiting N_2O consumption rather than production and enhancing N_2O emissions.

Comparing with SF, the application of did not significantly affect N_2O emissions produced via nitrification (Table 4), which is not consistent with the study by Fan et al. (2011). They reported that the use of noticeably enhanced nitrification rate relative to SF, for promoting the growth of ammonia-oxidizing bacteria (AOB). The enhanced growth of AOB is ascribed to the increase in soil pH caused by OF, as AOB prefer neutral to acidic pH (Liu et al., 2019). However, in this study, OF did not

Table 3
Soil characteristics in different treatments after four years of the maize-wheat rotation.

Treatment	pH	Organic C (g kg^{-1})	Total N (g kg^{-1})	Total P (g kg^{-1})	Total K (g kg^{-1})	Available P (mg kg^{-1})	Available K (mg kg^{-1})
CK	7.50 ± 0.15 a	11.69 ± 0.14 d	1.00 ± 0.03 c	0.73 ± 0.11 a	16.77 ± 0.15 a	64.36 ± 8.33 a	334.33 ± 37.3 a
TS	7.26 ± 0.12 ab	12.52 ± 0.44 d	1.33 ± 0.08 bc	0.82 ± 0.06 a	16.97 ± 0.03 a	53.4 ± 4.31 a	388.00 ± 41.20 a
LO	6.66 ± 0.19 c	13.00 ± 0.03 cd	1.46 ± 0.07 ab	0.79 ± 0.09 a	16.37 ± 1.11 a	59.92 ± 1.63 a	363.33 ± 45.86 a
MO	6.48 ± 0.04 c	15.92 ± 0.84 b	1.59 ± 0.19 ab	0.79 ± 0.06 a	17.02 ± 0.12 a	70.77 ± 6.71 a	368.00 ± 27.62 a
HO	6.50 ± 0.03 c	14.51 ± 0.49 bc	1.32 ± 0.03 bc	0.70 ± 0.09 a	17.15 ± 0.77 a	54.95 ± 4.69 a	346.33 ± 37.26 a
TO	6.97 ± 0.13 b	18.19 ± 0.92 a	1.72 ± 0.17 a	0.85 ± 0.04 a	17.12 ± 0.08 a	78.97 ± 10.14 a	399.00 ± 47.44 a

CK, no N fertilizer; TS, total synthetic N; LO, organic N: synthetic N = 1 : 2; MO, organic N: synthetic N = 1 : 1; HO, organic N: synthetic N = 2 : 1; TO, total organic fertilizer. Different letters mean significant difference at $p < 0.05$ between treatments.

Table 4
N_2O emission rates by denitrification and nitrification in wheat-growing (WS) and maize-growing (MS) soils under different treatments.

Treatment	N_2O emission rate (μg kg^{-1} d^{-1})			
	Denitrification in WS	Nitrification in WS	Denitrification in MS	Nitrification MS
CK	0.8 ± 0.2c	2.3 ± 0.7a	0.8 ± 0.2c	3.3 ± 0.2a
TS	1.0 ± 0.1c	1.2 ± 0.3 ab	0.9 ± 0.2c	2.8 ± 0.2a
LO	1.4 ± 0.2 b	1.2 ± 0.3 ab	1.8 ± 0.2 b	3.1 ± 0.3a
MO	1.5 ± 0.2 b	2.1 ± 0.4a	3.1 ± 0.3a	3.1 ± 0.7a
HO	2.2 ± 0.1a	1.0 ± 0.2 b	2.2 ± 0.4 b	3.9 ± 0.7a
TO	2.0 ± 0.1a	1.2 ± 0.1 ab	2.0 ± 0.3 b	3.0 ± 0.6a

CK, no N fertilizer; TS, total synthetic N; LO, organic N: synthetic N = 1 : 2; MO, organic N: synthetic N = 1 : 1; HO, organic N: synthetic N = 2 : 1; TO, total organic fertilizer. Different letters mean significant difference at $p < 0.05$ between treatments.

lead to a soil pH increase (Table 3), and thus might not enhance AOB growth. In addition, the concentration of NH_4^+ is another key factor affecting AOB growth in soils applied with OF (Kong et al., 2019). But in our case, as the capability of slow-release N for OF could be annually weakened, there was no significant difference in the concentration of NH_4^+ between SF-and OF- treated soils (Fig. 3a). Therefore, it may also not affect AOB growth and nitrification.

4.2. Intra- and interannual variation in N_2O emission

In this study, N_2O flux peaks took place after basal and supplemental fertilization (Fig. 1). Compared with supplemental application, basal fertilization induced the larger emission peaks, accounting for 52.63–73.98% of the mean cumulative emissions for the entire year (Table 1). This could be attributed to combination effects of fertilization and tillage. Unlike supplemental fertilizer, basal fertilizer inputs could provide substantial amount of both organic C and mineral N to denitrifying and nitrifying microbes. Furthermore, basal fertilizer was incorporated into soil by tillage. The tillage operation has been reported to be beneficial to microbial biomass and activity (Krauss et al., 2017). Thus, their combination effects could stimulate N_2O emissions more intensely than topdressing N alone.

Table 5
Response of N_2O emission rates to temperature fluctuations and N_2O increase per degree rise in temperature (IT) in wheat- and maize-growing soils under different treatments.

Season	Temperature (°C)	N_2O emission rate (μg kg^{-1} d^{-1})					
		CK	TS	LO	MO	HO	TO
Wheat	13.0	3.1 ± 0.4 b	1.8 ± 0.1c	1.9 ± 0.3 b	2.7 ± 0.2c	2.8 ± 0.4a	2.6 ± 0.5 b
	14.5	4.0 ± 0.2a	2.2 ± 0.1 b	2.2 ± 0.2 b	3.4 ± 0.2 b	3.1 ± 0.4a	3.2 ± 0.3 b
	17.0	4.4 ± 0.3a	2.6 ± 0.1a	3.1 ± 0.1a	4.2 ± 0.2a	3.0 ± 0.3a	3.9 ± 0.2a
	IT	0.5 ± 0.1	0.3 ± 0.1	0.5 ± 0.1	0.6 ± 0.0	0.1 ± 0.0	0.7 ± 0.1
Maize	24.0	3.4 ± 0.4 b	3.1 ± 0.2 b	3.6 ± 0.3 b	3.2 ± 0.1 b	3.7 ± 0.1 b	3.6 ± 0.2 b
	25.0	3.6 ± 0.6 b	3.3 ± 0.4 b	3.4 ± 0.3 b	3.0 ± 0.1 b	3.0 ± 0.2c	3.7 ± 0.4 b
	26.0	3.7 ± 0.3 b	3.2 ± 0.1 b	3.2 ± 0.4 b	3.3 ± 0.3 b	4.0 ± 0.5 b	3.8 ± 0.3 b
	29.0	4.6 ± 0.1a	3.7 ± 0.1a	4.8 ± 0.4a	4.8 ± 0.2a	5.0 ± 0.4a	4.4 ± 0.2a
	IT	0.2 ± 0.0	0.1 ± 0.0	0.3 ± 0.1	0.4 ± 0.1	0.4 ± 0.0	0.2 ± 0.0

CK, no N fertilizer; TS, total synthetic N; LO, organic N: synthetic N = 1 : 2; MO, organic N: synthetic N = 1 : 1; HO, organic N: synthetic N = 2 : 1; TO, total organic fertilizer. Different letters mean significant difference at $p < 0.05$ between temperatures.

There were great differences in size of N_2O flux peaks induced by basal application between years (Fig. 1). The inter-annual difference in peak size may be attributed to the obvious change of soil moisture and temperature. The highest N_2O peak sizes in the maize and wheat seasons were observed in 2015 and 2016, respectively (Fig. 1). The soil WFPS corresponding to the highest N_2O peak size of the maize season in 2015 was 45.8% (Fig. 2). Previous studies have demonstrated that 40–55% of soil WFPS is the most favorable for nitrification (Norton and Ouyang, 2019). Therefore, the WFPS (45.8%) in 2015 may be more conducive to N_2O emissions from nitrification than that in the other years (28.2–34.5%, Fig. 2). The soil temperature corresponding to the highest N_2O peak size in 2015 was 24 °C (Fig. 2). However, we did not found that the N_2O emission rate at 24 °C was significantly greater than those at the other temperatures (Table 5). These results indicate that compared with soil temperature, soil moisture may be the key factor controlling N_2O emissions in the maize season. By comparing the soil temperature and WFPS corresponding to the N_2O flux peaks in wheat seasons, we observed that soil temperature rather than WFPS in 2016 (17 °C) was obviously higher than those in the other years. Furthermore, the N_2O emissions across all treatments increased substantially with increase in soil temperature (Table 5). These findings indicate that soil temperature rather than moisture was the key factor determining the N_2O peak size in the wheat season. Therefore, choosing to apply base fertilizer to maize when soil is dry and applying fertilizer to wheat when temperatures are cool can help mitigate N_2O emissions.

We also found that the response of N_2O emission to temperature fluctuation was different among the fertilization treatments, supporting our second hypothesis. The increase rates of N_2O per unit temperature rise in all the OF treatments (except HO in wheat season) were significantly higher than that in the TS treatment (Table 5). Although the wheat-seasonal increase rate in HO was lower than that in TS, the maize-seasonal increase rate in HO was more than triple that in TS. Overall, as temperatures rise, N_2O emissions from OF-treated soils may be further enhanced than that from SF-treated soils in maize-wheat rotation systems. This supports earlier findings by Cui et al. (2016), showing that organic manure can strengthen the warming-induced increase of N_2O emissions by shifting community composition of *nirS*-encoding denitrifiers. Therefore, our results suggest that long-term OF substitution can increase the temperature sensitivity of N_2O emissions.

5. Conclusions

Compared with the application of SF alone, partial substitution of OFs can markedly reduce N_2O emissions in the first year. However, from the second year onwards, partial organic substitution did not reduce N_2O emissions, and even significantly enhanced them in the fourth year. The enhanced N_2O emissions from the OF-treated soils were mainly attributable to increased N_2O emissions produced via denitrification, accelerated OF decomposition, and lessened N_2O reduction. The application of basal fertilization induced the larger N_2O emission peaks than supplemental fertilization. There were great inter-annual differences in the sizes of N_2O flux peaks induced by basal fertilization. Soil moisture and temperature mediated the sizes in the maize and wheat seasons, respectively. Choosing to apply base fertilizer to maize when soil is dry and to wheat when temperatures are cool can help mitigate N_2O emissions. Long-term OF substitution enhance the temperature sensitivity of N_2O emissions and may further promote N_2O emissions as soils warm.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

CRediT authorship contribution statement

He Song: Conceptualization, Writing - original draft, Writing -

review & editing. **Jun Wang:** Investigation, Methodology, Writing - review & editing. **Kui Zhang:** Data curation. **Manyu Zhang:** Visualization. **Rui Hui:** Formal analysis. **Tianyi Sui:** Methodology. **Lin Yang:** Investigation. **Wenbin Du:** Project administration. **Zhaorong Dong:** Supervision, Funding acquisition.

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Appendix A. Supplementary data

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